Double Bracket Flow: Dynamical Systems that Diagonalize Matrices

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November 9, 2026

Abstract

This report reviews a result of Brockett [1]. For symmetric $n \times n$ matrices H and N, the *double bracket flow* $\dot{H} = [H, [H, N]]$, with $[A, B] = AB - BA$, is equivalent to a gradient flow on the orthogonal group. It preserves the eigenvalues of $H(0)$ and drives $H(t)$ to a diagonal matrix whose entries are ordered like the eigenvalues of N; the flow can therefore be regarded as an analog computer for diagonalizing, and simultaneously sorting the eigenvalues of, a symmetric matrix. We state the main results, give their proofs, and illustrate all of them on a 2×2 example that can be solved in closed form. We close with a brief comment on the recent use of this flow in quantum algorithms.

1 Introduction

Diagonalizing a symmetric matrix is usually done by discrete iterative algorithms. Brockett [1] observed that the same task can be carried out by a continuous-time dynamical system. Given a symmetric matrix Q and a fixed symmetric matrix N with distinct eigenvalues, consider

$$\dot{H} = [H, [H, N]], \qquad H(0) = Q. \tag{1.1}$$

The right-hand side is symmetric, so $H(t)$ remains symmetric, and, as shown in Section 2, $H(t)$ stays in the set of matrices orthogonally similar to Q, so its eigenvalues do not change. The results of Section 3 state that $H(t)$ converges to a diagonal matrix, that the limit is generically the one whose diagonal entries are ordered like the eigenvalues of N, and that this limit is the only asymptotically stable equilibrium.

Throughout we use the running example

$$Q_0 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad N = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \tag{1.2}$$

for which the flow can be solved explicitly (Section 4).

The report is organized as follows. Section 2 derives the flow from a gradient flow on $SO(n)$. Section 3 states and proves the four main results. Section 4 works through the 2×2 example. Section 5 discuss the derivation of the gradient flow from a geometric perspective. Section 6 comments on quantum extensions. The solution of the differential equation used in the example is given in the appendix.

2 The flow as a gradient flow on $SO(n)$

Let $SO(n)$ be the group of $n \times n$ orthogonal matrices with determinant 1, and let Q and N be fixed symmetric $n \times n$ matrices. For $\Theta \in SO(n)$ consider the function

$$\eta(\Theta) = \operatorname{tr}\big(Q\Theta N\Theta^T\big). \tag{2.1}$$

Brockett [2] showed that its gradient ascent flow on $SO(n)$ is

$$\dot\Theta = -\Theta N\Theta^T Q\Theta + Q\Theta N = -\Theta\big(N\Theta^T Q\Theta - \Theta^T Q\Theta N\big), \tag{2.2}$$

where the second form uses $\Theta\Theta^T = I$. Put $H = \Theta^T Q\Theta$. Then

$$\begin{aligned}\dot H &= -\Theta^T Q\big(\Theta N\Theta^T Q\Theta - Q\Theta N\big) - \big(\Theta^T Q\Theta N\Theta^T - N\Theta^T Q\big)Q\Theta \\ &= -HNH + H^2N - HNH + NH^2 \\ &= [H,[H,N]].\end{aligned} \tag{2.3}$$

Thus a solution of (2.2) with $\Theta(0) = I$ gives a solution $H = \Theta^T Q\Theta$ of (1.1). In particular $H(t)$ has the same eigenvalues as Q for all t, and $H(t)$ is diagonal exactly when $\Theta(t)$ diagonalizes Q.

3 Main results

3.1 Equilibria

An equilibrium of (2.2) is a point where its right-hand side vanishes.

Theorem 1 (Equilibrium points on $SO(n)$)**.** *Suppose that $Q = \Psi^T D_Q \Psi$ and $N = \Xi^T D_N \Xi$ with Ψ and Ξ orthogonal, $D_Q = \operatorname{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, and $D_N = \operatorname{diag}(\mu_1, \mu_2, \dots, \mu_n)$. Assume that $\lambda_1 > \lambda_2 > \cdots > \lambda_n$ and $\mu_1 > \mu_2 > \cdots > \mu_n$. Then the values of Θ such that $\Theta N\Theta^T Q\Theta - Q\Theta N$ vanish are the values of Θ such that*

$$\widetilde\Theta = \Psi\Theta\Xi^T \tag{3.1}$$

is the product ΠD of a permutation matrix Π and a diagonal square root of the identity, D. That is, $D = \operatorname{diag}(\pm1, \pm1, \dots, \pm1)$.

Proof. (See also Ref. [2].) We may assume Q and N are diagonal. The (i,j) entry of $NP - PN$ is $(n_{ii} - n_{jj})p_{ij}$, so $NP - PN = 0$ only if P is diagonal. Hence $N\Theta^T Q\Theta - \Theta^T Q\Theta N = 0$ only if $\Theta^T Q\Theta = D$ is diagonal. Then D has the same eigenvalues as Q, and $Q\Theta = \Theta D$, which reads

$$\lambda_i\theta_{ij} = \lambda_{\pi(j)}\theta_{ij}, \qquad 1 \le i, j \le n, \tag{3.2}$$

for some permutation π. Since the λ_i are distinct, either $i = \pi(j)$ or $\theta_{ij} = 0$. For fixed i there is exactly one j with $\pi(j) = i$, so each row of Θ has at most one nonzero entry; as the rows are unit vectors, that entry is ±1. The converse is immediate. □

The equilibria are therefore the ways of matching the eigenvectors of Q with those of N, up to order and sign.

3.2 Convergence

Theorem 2 (Convergence to a diagonal matrix)**.** *Let N be a real diagonal matrix with distinct eigenvalues and let $H(0)$ be symmetric. If $\dot{H} = [H,[H,N]]$, then $H(\infty) = \lim_{t\to\infty} H(t)$ exists and is diagonal.*

Proof. Since $[H,N]$ is skew-symmetric, $[H,[H,N]]$ is symmetric and the flow stays in the space of symmetric matrices. A direct calculation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(HN) = \operatorname{tr}\big(N[H,[H,N]]\big) = -\operatorname{tr}\big((HN-NH)^2\big) = \operatorname{tr}\big((HN-NH)(HN-NH)^T\big) \geq 0, \tag{3.3}$$

where the third equality uses that $HN - NH$ is skew-symmetric. So $\operatorname{tr}(HN)$ is nondecreasing; it is bounded because $H(t)$ stays on the compact set of matrices orthogonally similar to $H(0)$. Hence it converges and its derivative tends to zero. The derivative vanishes only if $HN = NH$, and since N has distinct eigenvalues this holds if and only if H is diagonal. Therefore $H(t)$ approaches the set of diagonal matrices with the eigenvalues of $H(0)$ on the diagonal; this set is finite, so $H(t)$ converges to one of its points. □

This theorem alone does not say *which* diagonal matrix is reached, that is, in which order the eigenvalues of $H(0)$ appear. That is the content of the next two theorems.

3.3 Selection of the limit

Theorem 3 (Generic limit)**.** *Let Q and N be diagonal with distinct eigenvalues. Except for initial points in a finite union of codimension-one submanifolds of $SO(n)$, every solution of* (2.2) *approaches a matrix ΠD, where D is a diagonal square root of I and Π is the permutation matrix that reorders the diagonal of Q so as to maximize*

$$\eta = \sum_{i=1}^{n} q_{\pi(i)\pi(i)}\, n_{ii}. \tag{3.4}$$

The proof is given in Ref. [2]. By the rearrangement inequality the maximum is attained when the diagonal entries of Q are arranged in the same order as those of N. So for almost all initial conditions the flow ends at the sorted arrangement; the exceptional initial conditions lie on the stable manifolds of the other equilibria.

3.4 Linear stability

Theorem 4 (Equilibria and linear stability)**.** *Let N be symmetric with distinct eigenvalues $\mu_1 > \cdots > \mu_n$, and let $H(0)$ be symmetric with distinct eigenvalues $\lambda_1 > \cdots > \lambda_n$. Then $\dot{H} = [H,[H,N]]$ has $n!$ equilibria, one for each ordering of the eigenvalues of $H(0)$ on the diagonal. At the equilibrium corresponding to a permutation π, the eigenvalues of the linearization are*

$$\big(\lambda_{\pi(i)} - \lambda_{\pi(j)}\big)\big(\mu_j - \mu_i\big), \qquad 1 \leq i < j \leq n. \tag{3.5}$$

In particular, all of them have negative real parts if and only if π is the identity, i.e. if the diagonal of the equilibrium is ordered like N.

Proof. As in the proof of Theorem **??** we may take N diagonal. By Theorem 2 every equilibrium is of the form $\operatorname{diag}(\lambda_{\pi(1)}, \dots, \lambda_{\pi(n)})$, which gives $n!$ equilibria. Linearizing $[H,[H,N]]$ about such a point, the off-diagonal entries obey

$$\dot{h}_{ij} = \big(\lambda_{\pi(i)} - \lambda_{\pi(j)}\big)\big(\mu_j - \mu_i\big) h_{ij}, \qquad i < j, \tag{3.6}$$

which are the stated eigenvalues. For $i < j$ we have $\mu_j - \mu_i < 0$, so every factor is negative exactly when $\lambda_{\pi(i)} > \lambda_{\pi(j)}$ for all $i < j$, that is, $\pi = \mathrm{id}$. □

Consequently the ordered diagonal matrix is the only asymptotically stable equilibrium of the double bracket flow; every other equilibrium has at least one unstable direction.

4 A worked example

We now apply the results to Q_0 and N in (1.2). The eigenvalues are $\lambda_1 = 3,\ \lambda_2 = 1$ for Q_0 and $\mu_1 = 2,\ \mu_2 = 1$ for N.

4.1 Solving the flow

Write $H(t) = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$. Then

$$[H,N] = \begin{pmatrix} 0 & -b \\ b & 0 \end{pmatrix}, \qquad [H,[H,N]] = \begin{pmatrix} 2b^2 & b(d-a) \\ b(d-a) & -2b^2 \end{pmatrix}, \tag{4.1}$$

so (1.1) becomes

$$\dot{a} = 2b^2, \qquad \dot{b} = b(d-a), \qquad \dot{d} = -2b^2, \tag{4.2}$$

with $a(0) = 2$, $b(0) = 1$, $d(0) = 2$. As shown in Appendix A,

$$a(t) = 2 + \tanh 2t, \qquad b(t) = \operatorname{sech} 2t, \qquad d(t) = 2 - \tanh 2t, \tag{4.3}$$

and therefore

$$H(t) = \begin{pmatrix} 2 + \tanh 2t & \operatorname{sech} 2t \\ \operatorname{sech} 2t & 2 - \tanh 2t \end{pmatrix} \longrightarrow \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \quad (t \to \infty). \tag{4.4}$$

The limit is diagonal, has the eigenvalues of Q_0, and is ordered like N, in agreement with Theorems 2 and 3. The other possible limit, $\operatorname{diag}(1,3)$, is not reached.

4.2 The accompanying rotation

Let $\Theta(t)$ be the rotation by angle $\theta(t)$, with $\Theta(0) = I$, such that $H = \Theta^T Q_0 \Theta$. Then

$$\Theta^T Q_0 \Theta = \begin{pmatrix} 2 + \sin 2\theta & \cos 2\theta \\ \cos 2\theta & 2 - \sin 2\theta \end{pmatrix}. \tag{4.5}$$

Comparing with the solution above gives $\sin 2\theta = \tanh 2t$ and $\cos 2\theta = \operatorname{sech} 2t$, hence $\tan\theta = \tanh t$ and

$$\theta(t) = \arctan(\tanh t) \longrightarrow \frac{\pi}{4}. \tag{4.6}$$

Thus

$$\Theta(\infty) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \qquad \Theta(\infty)^T Q_0 \Theta(\infty) = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.7}$$

4.3 Equilibria on $SO(2)$

By the reduction in the proof of Theorem **??** it suffices to consider the diagonalized matrix $Q = \mathrm{diag}(3,1)$, with N as before. For $\Theta(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ one finds from (2.2)

$$\dot{\Theta} = -2\sin\theta\cos\theta \begin{pmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}, \qquad \text{i.e.} \qquad \dot{\theta} = -\sin 2\theta. \tag{4.8}$$

The equilibria are $\theta = 0, \pi/2, \pi, 3\pi/2$:

$$\Theta_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \Theta_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \Theta_3 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \Theta_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{4.9}$$

With the permutation matrices $\Pi_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\Pi_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ these are

$$\Theta_1 = \Pi_1\,\mathrm{diag}(1,1), \quad \Theta_3 = \Pi_1\,\mathrm{diag}(-1,-1), \quad \Theta_2 = \Pi_2\,\mathrm{diag}(1,-1), \quad \Theta_4 = \Pi_2\,\mathrm{diag}(-1,1), \tag{4.10}$$

as asserted by Theorem **??**. (Only the products ΠD, not Π_2 itself, need lie in $SO(2)$.) Since $\dot{\theta} = -\sin 2\theta$, the points $\theta = 0, \pi$ are stable and $\theta = \pi/2, 3\pi/2$ are unstable.

4.4 Selection of the limit

The two permutations $\pi_1 = \mathrm{id}$ and $\pi_2 = (1\,2)$ give

$$\eta_1 = q_{11}n_{11} + q_{22}n_{22} = 3\cdot 2 + 1\cdot 1 = 7, \qquad \eta_2 = q_{22}n_{11} + q_{11}n_{22} = 1\cdot 2 + 3\cdot 1 = 5. \tag{4.11}$$

Since $\eta_1 > \eta_2$, Theorem 3 predicts the limit $\mathrm{diag}(3,1)$, as found above. The four rotations $\Theta_1, \ldots, \Theta_4$ correspond to only two matrices $H = \Theta^T Q \Theta$: Θ_1, Θ_3 give $H_1 = \mathrm{diag}(3,1)$ and Θ_2, Θ_4 give $H_2 = \mathrm{diag}(1,3)$.

4.5 Linear stability

By Theorem 4 there are $2! = 2$ equilibria, H_1 and H_2, and the single off-diagonal perturbation has rate $\rho_\pi = (\lambda_{\pi(1)} - \lambda_{\pi(2)})(\mu_2 - \mu_1)$.

- At H_1 ($\pi = \mathrm{id}$): $\rho = (3-1)(1-2) = -2$, so $h_{12}(t) \approx h_{12}(0)e^{-2t} \to 0$. The equilibrium is asymptotically stable.
- At H_2 ($\pi = (1\,2)$): $\rho = (1-3)(1-2) = 2$, so $h_{12}(t) \approx h_{12}(0)e^{2t}$. Any nonzero perturbation grows exponentially, so the equilibrium is unstable.

This agrees with the sign of $\dot{\theta} = -\sin 2\theta$ found above.

5 Geometric Derivation

5.1 The Isospectral Manifold

Our objective is to construct a dynamical system that diagonalizes a real symmetric matrix H_0, while preserving its eigenvalues throughout the evolution.

By the spectral theorem, orthogonal similarity transformations preserve the eigenvalues of a symmetric matrix. We therefore consider

$$H = Q^T H_0 Q, \qquad Q \in SO(n), \tag{5.1}$$

where

$$SO(n) = \{Q \in GL(n, \mathbb{R}) : Q^T Q = I,\ \det Q = 1\}.$$

The group $SO(n)$ is a Lie group, and its action on symmetric matrices generates the set

$$\mathcal{O}(H_0) = \{Q^T H_0 Q : Q \in SO(n)\}. \tag{5.2}$$

This set is called the *isospectral manifold*. It contains precisely the symmetric matrices with the same eigenvalues as H_0.

Thus, instead of searching over the entire space of symmetric matrices, we seek a dynamical system evolving on $\mathcal{O}(H_0)$.

5.2 The Tangent Space

To construct a dynamical system on this manifold, we first determine its admissible tangent directions.

The Lie algebra of $SO(n)$ is

$$\mathfrak{so}(n) = \{\Omega : \Omega^T = -\Omega\}. \tag{5.3}$$

For any $\Omega \in \mathfrak{so}(n)$, the exponential map generates a curve

$$Q(t) = Qe^{t\Omega}$$

on $SO(n)$.

Consequently,

$$H(t) = e^{-t\Omega} H e^{t\Omega}$$

is a curve on the isospectral manifold passing through H.

Differentiating at $t = 0$ gives

$$\dot{H}(0) = -\Omega H + H\Omega = [H, \Omega]. \tag{5.4}$$

Therefore,

$$T_H \mathcal{O}(H_0) = \{[H, \Omega] : \Omega \in \mathfrak{so}(n)\}. \tag{5.5}$$

This shows that an isospectral evolution must take the form

$$\dot{H} = [H, \Omega(H)]. \tag{5.6}$$

The remaining problem is to determine $\Omega(H)$ so that the evolution approaches a diagonal matrix.

5.3 Constructing a Gradient Flow

To select a direction of evolution, we introduce a fixed diagonal matrix

$$N = \operatorname{diag}(n_1, \dots, n_n),$$

with distinct diagonal entries, and define the objective function

$$F(H) = \operatorname{tr}(HN). \tag{5.7}$$

The motivation is that, on the isospectral manifold, the critical points of this function satisfy $[H, N] = 0$. Since N has distinct diagonal entries, this condition is equivalent to H being diagonal.

We therefore seek a gradient ascent flow of F.

To define a gradient, we need a Riemannian metric. On $SO(n)$, we choose the metric

$$g_Q(Q\Omega_1, Q\Omega_2) = -\operatorname{tr}(\Omega_1\Omega_2). \tag{5.8}$$

Now regard the objective as a function of Q:

$$f(Q) = F(Q^T H_0 Q) = \operatorname{tr}(Q^T H_0 Q N). \tag{5.9}$$

For a tangent direction $Q\Omega$, we have

$$\begin{aligned} df_Q(Q\Omega) &= \operatorname{tr}(N[H, \Omega]) && (5.10)\\ &= \operatorname{tr}([N, H]\Omega) && (5.11)\\ &= -\operatorname{tr}([H, N]\Omega). && (5.12) \end{aligned}$$

By definition, the Riemannian gradient satisfies

$$g_Q(\operatorname{grad} f, Q\Omega) = df_Q(Q\Omega) \tag{5.13}$$

for every $\Omega \in \mathfrak{so}(n)$.

Comparing both expressions, we obtain

$$\operatorname{grad} f(Q) = Q[H, N]. \tag{5.14}$$

Hence the gradient ascent equation on $SO(n)$ is

$$\dot{Q} = Q[H, N]. \tag{5.15}$$

Finally, differentiating $H = Q^T H_0 Q$ gives

$$\begin{aligned} \dot{H} &= \dot{Q}^T H_0 Q + Q^T H_0 \dot{Q} && (5.16)\\ &= -[H, N]H + H[H, N] && (5.17)\\ &= [H, [H, N]]. && (5.18) \end{aligned}$$

Thus, the double bracket flow arises naturally from gradient ascent on the orthogonal group, subject to the constraint of preserving the eigenvalues of H.

Furthermore,

$$\frac{d}{dt} F(H(t)) = \|[H, N]\|_F^2 \geq 0, \tag{5.19}$$

so the objective increases monotonically and is stationary precisely when H commutes with N.

Therefore, the geometric construction explains both the isospectral property and the tendency toward diagonalization.

6 Discussion and outlook

The double bracket flow diagonalizes a symmetric matrix and sorts its eigenvalues without any explicit eigenvalue computation. Double-bracket flows have recently inspired quantum algorithms for diagonalizing Hamiltonians [3]; the work of Prof. Marek Gluza (SPMS) in this direction motivated this report. In these algorithms the continuous flow is approximated by a sequence of unitary quantum-circuit operations that successively suppress off-diagonal terms and prepare approximate eigenstates. The framework may also be useful for ground-state preparation, although the required circuit depth can grow rapidly with the number of iterations.

A Solution of the differential equation (4.2)

We solve (4.2) with $a(0) = 2$, $b(0) = 1$, $d(0) = 2$. First, $\frac{\mathrm{d}}{\mathrm{d}t}(a + d) = 2b^2 - 2b^2 = 0$, so $a + d = 4$. Define $x = \frac{1}{2}(a - d)$. Then $\dot{x} = 2b^2$, and since $d - a = -2x$, the equation for b reads $\dot{b} = -2xb$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(x^2 + b^2\big) = 2x\dot{x} + 2b\dot{b} = 4xb^2 - 4xb^2 = 0, \tag{A.1}$$

and with $x(0) = 0$, $b(0) = 1$ we get $x^2 + b^2 = 1$. Substituting $b^2 = 1 - x^2$ gives $\dot{x} = 2(1 - x^2)$, and separating variables,

$$\frac{\mathrm{d}x}{1 - x^2} = 2\,\mathrm{d}t \quad \Longrightarrow \quad x(t) = \tanh 2t. \tag{A.2}$$

Since $b(0) > 0$, $x^2 + b^2 = 1$ yields $b(t) = \operatorname{sech} 2t$. Finally $a = 2 + x$ and $d = 2 - x$ give (4.3).

References

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